Chapter 4: Coupled Microstrip Line

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Note: This chapter presents research results on techniques to suppress far-end crosstalk (FEXT) in coupled microstrip lines.

1.1 4.1 Literature Survey *(pp.37)*

Several existing methods for suppressing FEXT in coupled microstrip lines:

Method	FEXT Reduction	Drawback
Guard trace [1]	Slight	Minimal improvement
Grounded guard trace [2]	Efficient	Limited routing due to vias
Grounded guard trace + dielectric overlay [3]	Efficient, only 2 vias	Extra dielectric overlay cost
Serpentine guard trace [4]	Efficient, no vias/overlay	Large area
Alternately open-circuited stubs [5]	Efficient, less area	Large reflection

Motivation for this chapters approach: eliminate the need for dielectric overlay, vias, serpentine guard trace, AND reduce reflection using a **decoupling capacitor** method.

1.2 4.2 Baseline: Coupled Microstrip Line Parameters *(p.9)*

Substrate: FR4, $\varepsilon_r = 4.4$, $\tan\delta = 0.02$

Dimensions: $w = 1.42$ mm, $s = 0.7$ mm, $h = 0.8$ mm, $\ell = 40$ mm

Equivalent circuit per unit length: series self-inductance L_s, mutual inductance L_m, shunt self-capacitance C_s, mutual capacitance C_m.

1.3 4.3 Even- and Odd-Mode Equivalent Circuits *(p.10)*

Mode	Series inductance	Shunt capacitance	Time delay
Even	$L_s + L_m$	C_s	$\mathrm{TD}_{\mathrm{even}} = \ell\sqrt{(L_s + L_m)C_s}$
Odd	$L_s - L_m$	$C_s + 2C_m$	$\mathrm{TD}_{\mathrm{odd}} = \ell\sqrt{(L_s - L_m)(C_s + 2C_m)}$

(chap4.pdf, p.10)

Key finding: $\mathrm{TD}_{\mathrm{even}} \neq \mathrm{TD}_{\mathrm{odd}}$ in a microstrip. The time difference is:

$$T_d = \mathrm{TD}_{\mathrm{even}} - \mathrm{TD}_{\mathrm{odd}}$$

This difference T_d is **the root cause of FEXT** in a microstrip line (unlike stripline, where $T_d = 0$).

1.4 4.4 FEXT and NEXT Ideal Coupled Line *(pp.1113)*

Even/odd mode decomposition: A direct excitation on the active line decomposes into even-mode excitation (both lines driven $+V_s/2$) and odd-mode excitation (active $+V_s/2$, quiet $-V_s/2$).

For the baseline (ideal) coupled line: - $\mathrm{TD}_{\mathrm{even}} = 253.614$ ps, $\mathrm{TD}_{\mathrm{odd}} = 229.202$ ps, $T_d = 24.412$ ps - **FEXT peak** = 0.076 V - **NEXT peak** = 0.019 V

FEXT = $V_{4o}(t) + V_{4e}(t)$ the even and odd mode contributions arrive at different times due to T_d, producing a pulse.

1.5 4.5 FEXT/NEXT Estimation Equations (Baseline) *(p.14)*

NEXT peak (at $t = t_r$):

$$V_2(t_r) = V_{2e}(t_r) + V_{2o}(t_r)$$

$$V_{2e}(t_r) = \frac{V_s}{2}\left(\frac{Z_{\mathrm{even}}}{Z_{\mathrm{even}} + Z_0}\right) - \frac{V_s}{2} \cdot \frac{Z_{\mathrm{even}} - Z_0}{Z_{\mathrm{even}} + Z_0}$$

FEXT peak (at $t = \mathrm{TD}_{\mathrm{even}}$):

$$|V_4(\mathrm{TD}_{\mathrm{even}})| = V_{\max} \cdot \min\left(\frac{T_d}{t_r}, 1\right)$$

where $V_{\max} = \frac{Z_0}{Z_{\mathrm{even}} + 2Z_0 + Z_{\mathrm{odd}}} \cdot V_s$
(chap4.pdf, p.14)

1.6 4.6 Coupled Microstrip with Front-End Decoupling Capacitor *(pp.2233)*

1.6.1 Topology *(p.22)*

A capacitor C_f is placed at the **near-end (source end)** of the quiet line.

1.6.2 Design Concept *(p.24)*

The goal is to make $\mathrm{TD}_{\mathrm{even}} = \mathrm{TDc}_{\mathrm{odd}}$ by adding C_f:

The odd-mode time delay with C_f added becomes:

$$\mathrm{TDc}_{\mathrm{odd}} = \ell\sqrt{(L_s - L_m)\left(C_s + 2C_m + \frac{2C_f}{\ell}\right)}$$

(chap4.pdf, p.23)

Setting $T_d = \mathrm{TDc}_{\mathrm{odd}} - \mathrm{TD}_{\mathrm{even}} = 0$ gives the required C_f.

Example: $T_d = 24.412$ ps $C_f = 0.6$ pF (ideal line) *(p.25)*

1.6.3 Effect on FEXT and NEXT *(pp.2627)*

Parameter	Baseline	With C_f
T_d (ideal)	24.412 ps	0
FEXT peak (ideal)	0.076 V	0.027 V
NEXT peak (ideal)	0.019 V	0.080 V

Trade-off: Adding C_f suppresses FEXT but **increases NEXT** (because the front-end capacitor interacts with the odd-mode wave).

1.6.4 Lossy line results *(pp.3233)*:

- FEXT: simulation 0.0257 V, measurement 0.0268 V (baseline was 0.063/0.055 V)
- NEXT: simulation 0.0710 V, measurement 0.0636 V (baseline was 0.018/0.016 V)

1.7 4.7 Eye Diagram Comparison *(pp.3436)*

Simulation with pseudo-random bit sequence (PRBS):

Bit Rate	Baseline Eye Height	With C_f Eye Height
2 Gbps	97.2%	97.2%
4 Gbps	95.2%	88.8%
8 Gbps	94.0%	86.0%
16 Gbps	79.6%	64.4%

Conclusion: Eye diagram performance degrades with front-end C_f at high bit rates. *(p.36)*

1.8 4.8 Distributed Decoupling Capacitors *(pp.3940)*

Instead of a single lumped C_f, distribute ΔC_f capacitors **along the line**.

Number of caps	Spacing	ΔC_f per cap	FEXT
4 ($\lambda/2$)	every $\lambda/2$	0.125 pF	better
8 ($\lambda/4$)	every $\lambda/4$	0.0625 pF	better
20 ($\lambda/10$)	every $\lambda/10$	0.025 pF	0

(chap4.pdf, p.39)

S-parameter improvement: Distributed caps restore the good transmission coefficient (S_{13}) and reduce reflections (S_{11}) that were degraded by the lumped front-end cap. *(pp.4142)*

1.9 4.9 Summary *(p.43)*

Method	FEXT (V meas/sim)	NEXT (V meas/sim)
Conventional	0.055 / 0.063	0.016 / 0.018
Front-end C_f	0.027 / 0.026	0.064 / 0.071
Distributed C_f	/ 0	/ 0.028

- Front-end C_f: reduces FEXT but increases NEXT and degrades eye at high bit rate.
- Distributed C_f: restores eye quality and achieves near-zero FEXT with acceptable NEXT.